

Short-term electric load forecasting for buildings using logistic mixture vector autoregressive model with curve registration

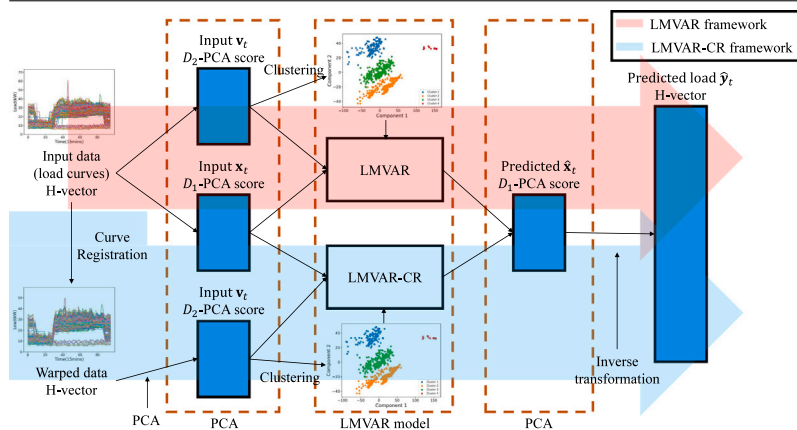
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GRAPHICAL ABSTRACT



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ABSTRACT

This paper proposes a day-ahead electric load forecasting model for buildings where daily load curves follow a few distinctive patterns. A pattern lasts for several days before changing into another. We particularly explore the problem that the day-ahead curve mostly depends on the load pattern history and is relatively insensitive to external environments such as weather conditions. The problem considers clusters for daily curve patterns and a day-ahead electric curve forecast from previous electric load and pattern history. We propose a model called the logistic mixture vector autoregressive model (LMVAR) that combines both clustering and forecasting in a single model through the expectation–maximization (EM) algorithm. To improve internal clustering performance, we apply the curve registration technique to the model. We test two models (the models with/without curve registration) with electric load data sets collected from a library and a grocery store. We then compare them with existing forecasting methods such as persistence, sequence-to-sequence long short-term memory (S2S LSTM), seasonal autoregressive (SAR), multiple-output support vector machine (M-SVM), multilayer perceptron (MLP), and a cluster-based model. The result shows that the proposed models outperform the benchmark methods.

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1. Introduction

The global electric demand has risen by 160% over the last two decades, and IEA [1] projects its growth to be 2.1% per year until 2040. The rapid increase in electricity demand necessitates the power grid's flexibility to reduce electricity wastage. So, the smart grid has been growing as the next-generation power grid as a viable solution. The smart grid improves electric systems' efficiency by exchanging information among its participants—e.g., utility companies, power generators, and consumers—in real-time using information and communication technology (ICT). The critical factor in operating a smart grid is balancing between demand and fluctuating power supply from renewable energy sources, and optimal strategies in operating energy storage systems (ESS) and demand-response do heavily rely on accurate forecasting skills both electric load and generation [2].

Notably, the short-term day-ahead load forecasting becomes more important as the electric market often requires a day-ahead bid to prepare for the next-day operation [3]. For each day, utility companies make a forecast on the next day's demand for electricity, which becomes the base for making a bid. Failure in load forecasting may cause an imbalance between demand and supply, which imposes an unexpected loss to utility companies and often ruins the entire grid, such as power outages. In forecasting short-term loads, we notice that the building sector takes a significant portion of global energy consumption. According to International Energy Outlook [4], the building sector consumed 20% of the global energy consumption in 2018, and it will increase to 22% until 2050. Therefore, the building energy forecasting for a short-term period is an essential problem for energy markets as well as the operations of smart grids. In this regard, this paper is focused on a day-ahead electric load forecasting for buildings.

1.1. Literature review

There have been many studies in the literature regarding general building energy demand forecasting; we refer the reader to the recent review papers [5,6] and the references therein. This section, instead, conducts a literature review for building electric load forecasting, focusing on a methodological perspective since we make a methodological contribution to the literature improving forecasting accuracy. The forecasting methods for building electric loads can be classified into three categories: physics-based, machine learning-based, and statistical approaches. Our proposed method belongs to the statistical approach, and we will compare the forecasting performance of our method with that of other approaches in the rest of the paper.

There are several physical simulation tools for building energy such as EnergyPlus [7], ESP-r [8], and TRNSYS [9]. Li and Wen [10] developed the on-line building energy forecasting model using the results of EnergyPlus simulation. They obtained building operation data from EnergyPlus simulation and represented a mathematical model related to a physical system. Due to the use of the simulation tools, the physics-based approach is often difficult to obtain mathematical expressions of the energy mechanisms of various buildings.

The machine learning approach to building electric load forecasting includes support vector machine (SVM) [11–15], artificial neural network (ANN) [16–22] and other machine learning techniques [23–25]. In the machine learning approach, feature extraction and model architecture affect the performance of short-term prediction. Paudel et al. [12] improved the SVM model for building thermal energy demand prediction by extracting features with clustering of the building operation profile, e.g. office buildings and residences buildings with climatic variables in weather data. Chen and Tan [11] employed 3-layer multi-resolution wavelet decomposition (MWD) and divided the original time series into several parts as features.

Fan et al. [16] assessed various deep recurrent neural network (RNN) architectures consisting of high-level and low-level characteristics for building energy prediction. The high-level characteristics

are three inference approaches: recursive, direct, and multi-input-and-multi-output (MIMO). The low-level characteristics include the types of RNN and the usage of a bidirectional model and other types of neural network architectures such as the convolutional neural network (CNN). Chitalia et al. [22] compared variations of long short-term memory models (LSTMs) with or without direction, attention, and CNN. Marino et al. [21] implemented a sequence-to-sequence (S2S) LSTM for building energy load forecasting. The prediction performance depends on how to utilize large-size data, extract features, and construct a model architecture. Despite the promising result, however, the machine learning approach has issues with large data collection and computation; it may not be easy to collect a large amount of data due to cost or environmental restrictions. The estimation of a large number of parameters imposes a computational burden.

The statistical approach requires relatively less amount of data than machine learning does. It uses regression and time series models [26–29]. Traditionally, autoregressive (AR) models have been implemented in time series prediction with only historical data. When external data is available, autoregressive with exogenous variables (ARX) [29] models and multiple linear regression [28] are used. Yun et al. [29] proposed a time-varying ARX model for a one-hour-ahead building thermal load forecast. Considering the seasonality of the electric load time series, Tarsitano and Amerise [26] proposed a two-stage seasonal autoregressive integrated moving average model with the exogenous variables (SARIMAX). Cai et al. [30] and Kim et al. [31] compared neural network-based models (ANN, RNN/CNN, and Nonlinear autoregressive with exogenous variable (NARX)) with time series models such as SARIMAX. The result revealed that neural network model dominated time series models in most cases. Time series models, however, have fewer parameters and better computational efficiency. So, they are suitable for forecasting when the size of the available data is limited.

For forecasting a day-ahead (or multi-step) time series, most models take either an iterative, a direct, or a multi-input multi-output (MIMO) approach [32]. The iterative method is to iterate several prediction steps using the previously predicted values as inputs: e.g., AR and autoregressive integrated moving average (ARIMA). This approach is not appropriate when the number of steps to forecast is large because the forecast error accumulates over iterations. The direct approach is that multiple models correspond to multiple ahead steps, so each model predicts each step of loads. This method has weaknesses to capture the dependence among models. The MIMO strategy is that a single model predicts a day-ahead electric load consisting of multiple values simultaneously with multiple inputs overcoming the above two approaches' limitations. Our proposed models also take the MIMO approach, and we choose benchmark models taking the same approach.

1.2. Motivation and contribution

As in the literature review, there are many methodologies we can use for building load forecasting. However, as we investigate building electric load data, we notice a couple of characteristics that existing approaches may not address fully, which motivates us to develop a new methodology for improving estimation accuracy.

First, the size of the data may not be large enough. Collecting and storing historical electricity usage data is costly. So, historical data is likely stored and managed just for a limited amount of time. Machine learning models such as SVM, neural network, and S2S LSTM require a large amount of data. Estimation using a small amount of data makes it difficult to find patterns and causes overfitting because its complexity is more than necessary. These overfitted models may show poor forecasting performance since they are hard to respond to out-of-sample.

Second, building load data shows distinct patterns. The factor that distinguishes load patterns is jumps in load. For example, a university library has a sudden decrease in electric load at 10 pm during a vacation period while the load drops at 2 am during a mid-term period.

The other characteristic is that jumps in load curves of the same pattern have noise across time due to human behavior such as library staff's daily variation in work hours or other external environments. This kind of noise across time is called *phase variability*. We note that Section 2 will detail the term *jump* and *pattern* in electric load curves, and Section 3.3 will describe the *phase variability*. Statistical approaches such as SAR can consider seasonality for the time series, but not the variation between load patterns. Other approaches in the literature address this kind of pattern *only implicitly* due to their general applicability. The cluster-based models such as [25] can consider daily load patterns *explicitly*. They, however, usually cluster the daily load curves and predict a load pattern and total load for the next day step by step; clustering and forecasting are *separated*.

To address the distinct characteristics of building electric load data that existing approaches may not do well, we propose a new model that works with *small-size data* and considers the *distinct patterns explicitly into a single unified* forecasting model. To do so, we propose the logistic mixture vector autoregressive model (LMVAR) that integrates pattern identification and forecasting with the MIMO approach: Wong and Li [33] first proposed a similar logistic mixture autoregressive model with a scalar input, and we extend the model for a vectorial input. The LMVAR model uses the daily load curve as the vectorial input, and we can link a relationship between the variations within a day and between load patterns of days. The pattern identification and model estimation can be iteratively performed through the expectation–maximization algorithm (EM).

The important thing in the proposed model is finding clear patterns. However, daily load curves have *phase variability* that hinders the pattern clustering even if the shapes of the curves are the same. Curve registration aligns the curves on their domain (time horizon), i.e., remove phase variability. It is well studied in functional data analysis literature [34,35]. So, expecting the enhancement of the clustering performance, we integrate curve registration into the LMVAR, referred to as LMVAR-CR. We will explain in detail in Section 4.

In this regard, we can summarize our contributions as follows:

- We develop a time series-based model that works well for relatively small-size data. The proposed model combines the mixing proportion of explicit daily load patterns and the vector autoregressive forecasting model in a single model with the MIMO approach.
- The proposed model is called the LMVAR, extending the LMARX model to integrate multiple vector autoregressive models with the mixing proportion expressed in logistic terms for short-term electric load forecasting.
- We incorporate the concept of phase variability into the proposed model by curve registration to effectively improve forecasting accuracy.

1.3. Organization of our paper

The organization of the paper is as follows: Section 2 describes the problem of our interest. Section 3 proposes the LMVAR model and discusses possible improvement of internal clustering in the model explaining the concept of phase variability and curve registration. Section 4 describes how we incorporate curve registration into the proposed model. Section 5 provides case studies using the real data set from library and grocery buildings and shows the effectiveness of the proposed model. Section 6 summarizes the paper and makes a conclusion.

2. Problem description

We consider a day-ahead electric load forecasting problem. An electric load is an electric demand at a specific time, and a day-ahead electric load forecasting is the task of predicting the electric loads

over the next 24 h. Suppose that electric loads are measured H times every day with a uniform time interval of length m . In our motivating example, $H = 96$ and $m = 15$ minutes. Let $\mathbf{y}_t = (y_{1,t}, y_{2,t}, \dots, y_{H,t})$ represent the electric load vector (or curve) consisting of H load measurements taken on day t where $y_{h,t}$ is the h th load measurement on day t . We define a pattern of curves to be a family of $\mathbf{y}_t$ that have similar variations of the values in $\mathbf{y}_t$. As illustrated in Fig. 1(b), there exist four distinct patterns of the load vector $\mathbf{y}_t$. Patterns in Fig. 1(b) exhibit sudden electric load changes at different times—we call those changes *jumps*. For example, pattern 1 in Fig. 1(b) has jumps around $h = 30$ and $h = 88$ while pattern 4 in Fig. 1(b) has no jumps.

The load vectors often follow one pattern for several successive days before the pattern changes to another. As illustrated in Fig. 2(b), the load pattern stays at pattern 1 for 17 days, moves to pattern 4 that lasts for four days, and returns to pattern 1. When the same pattern repeatedly occurs for several days, the load vectors on those days appear similar but not exactly the same, showing some temporal correlations. The temporal correlation in the load vectors from a pattern can be represented as a time series model. When the pattern changes to another, the load vectors follow another time series model that represents the new pattern. The switching among different time series models can be naturally modeled as a mixture of time series models.

We aim to develop a new approach to forecast the day-ahead electric load profile $\mathbf{y}_{t+1}$ when $\mathbf{y}_t$ follows a mixture of time series model. Fig. 3 illustrates the entire framework of our proposed approach. We propose a mixture model called logistic mixture vector autoregressive (LMVAR) model considering the clustering of curve patterns and forecasting a curve simultaneously using the principal component analysis (PCA) scores. We apply curve registration to the LMVAR model so that matching jumping moments results in improving internal clustering and forecasting accuracy (the LMVAR-CR model). More details of the LMVAR and the LMVAR-CR models will be described in Sections 3 and 4.

3. Logistic mixture vector autoregressive (LMVAR) model

This section describes the forecasting model, which is red in Fig. 3, to predict $\mathbf{y}_{t+1}$ given all past observations of the electricity load vectors $\mathcal{F}_t = \{\mathbf{y}_s | 1 \leq s \leq t\}$. Since $\mathbf{y}_t$ is quite high dimensional, we consider a low-dimensional projection of $\mathbf{y}_t$. We select two projected features— $\mathbf{w}_t$ for clustering load vectors and $\mathbf{x}_t$ for forecasting the overall load vector—incorporating the pattern-to-pattern variation and the within-pattern variation. For $\mathbf{x}_t$, we take the D_1 PCA scores of $\mathbf{y}_t$ that are along the D_1 maximum variance directions to make more distinction among load vectors. For $\mathbf{w}_t$, we take two different approaches. The first one is to take a few leading PCA scores as in the feature selection for the clustering. Specifically, we simply take the D_2 PCA scores of $\mathbf{y}_t$ in this section. We will discuss some limitations of this approach in Section 3.3 and will propose another approach, that is blue in Fig. 3, in Section 4 as an advancement of our initial proposal.

3.1. Model

Suppose that a daily electric load curve comes from one of K unknown patterns. We would like to identify the K unknown patterns and exploit the patterns and their temporal changes in forecasting the daily curves. The number K may be unknown in practice. For such cases, one can estimate the number K by using a clustering method such as the Bayesian Gaussian mixture clustering method (BGMC, [36,37]). Below is the description of a mixture of K autoregressive models to represent the time series of daily electric load curves.

An electric load curve for day t is converted to the D_1 PCA scores $\mathbf{x}_t \in \mathbb{R}^{D_1}$. When $\mathbf{x}_t$ comes from the k_t th pattern, the conditional distribution of $\mathbf{x}_t$ given $k_t = k$ is assumed to follow a multivariate Gaussian distribution,

$$\mathbf{x}_t | k_t = k, \mathcal{F}_{t-1} \sim \Phi(\mathbf{x}_t; \boldsymbol{\mu}_{k,t}, \sigma_k^2 \mathbf{I}). \quad (1)$$

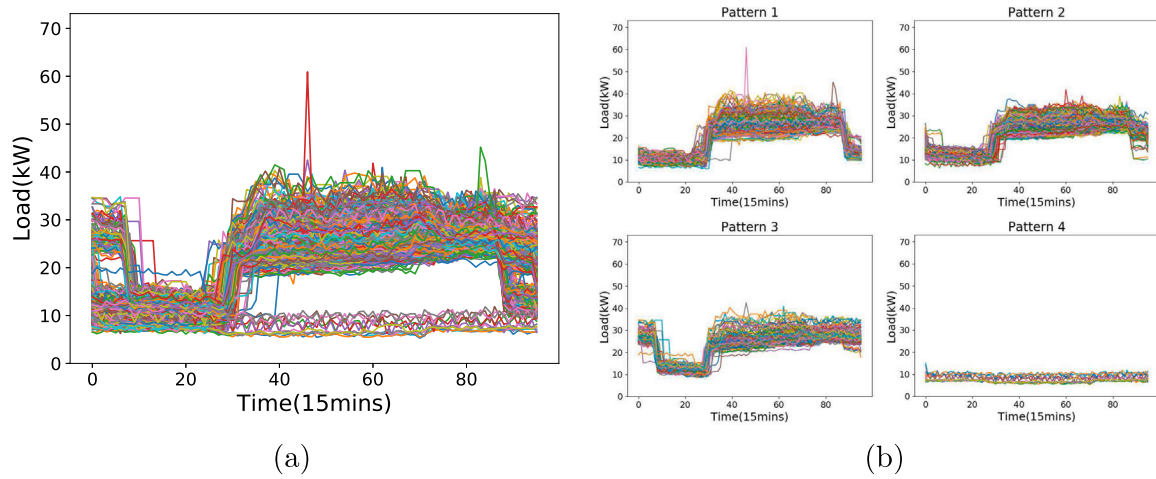


Fig. 1. (a): Daily electric load profiles for a library building (b): Electric load curves for each pattern.

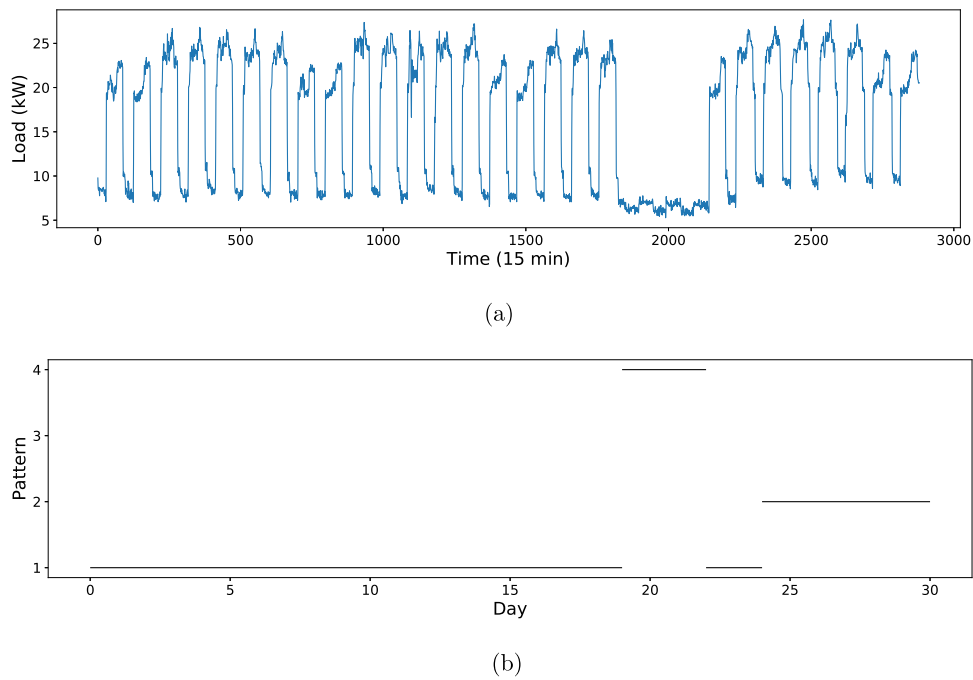


Fig. 2. (a): Electric load from Jan 26, 2018 to Feb 24, 2018 (b): Load patterns from Jan 26, 2018 to Feb 24, 2018.

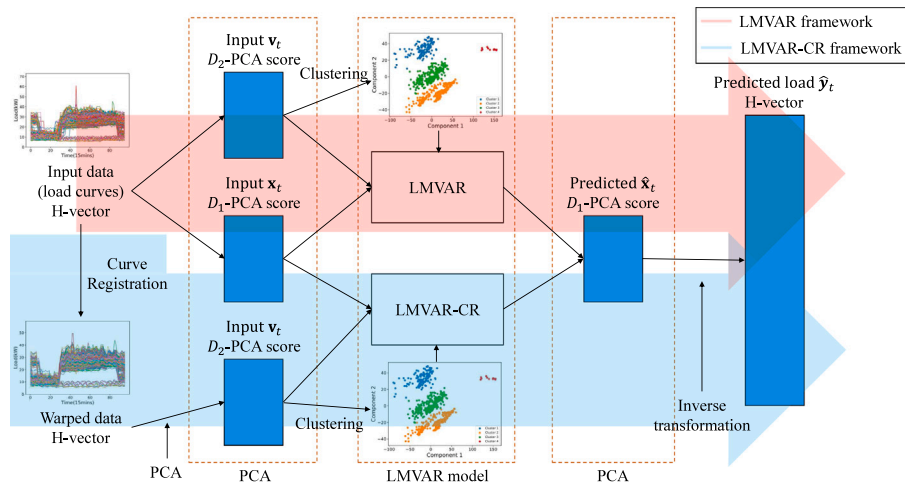


Fig. 3. Entire framework of our proposed models.

where $\Phi(\cdot; \mu, \Sigma)$ is the cdf of a multivariate Gaussian distribution with the mean vector μ and the covariance matrix Σ , and σ_k^2 is the variance of load curves within pattern k , I is an identity matrix. The mean vector $\mu_{k,t}$ is time-varying, which is modeled as the following autoregressive model,

$$\mu_{k,t} = \sum_{j=1}^{p_k} \varphi_{k,j} \mathbf{x}_{t-j},$$

where p_k is the time lag of load pattern k , and $\varphi_{k,j}$ is a VAR coefficient of load pattern k on day $t-j$.

The pattern index k_t is modeled as a multinomial random variable with its probability distribution,

$$P(k_t = k | \mathbf{w}_{(t-1):q}, \mathcal{F}_{t-1}) = \alpha_{k,t}, \quad (2)$$

where $\alpha_{k,t}$ is a logistic link function that relates the features for the past q daily load curves $\mathbf{w}_{(t-1):q} = (\mathbf{w}_{t-q}^T, \dots, \mathbf{w}_{t-1}^T)^T$ to the pattern probability through a $(D_2 \cdot q)$ -dimensional weight vector β_k for pattern k , $k \in \{1, \dots, K\}$,

$$\alpha_{k,t} = \frac{\exp(\mathbf{w}_{(t-1):q}^T \beta_k)}{\sum_{j=1}^K \exp(\mathbf{w}_{(t-1):q}^T \beta_j)} \quad \text{for } k = 1, \dots, K.$$

Accordingly, the joint distribution of $\mathbf{x}_t$ and k_t can be induced by taking the product of (1) and (2):

$$P(\mathbf{x}_t, k_t = k | \mathcal{F}_{t-1}) = \alpha_{k,t} \Phi(\mathbf{x}_t; \mu_{k,t}, \sigma_k^2 I).$$

The marginal distribution $p(\mathbf{x}_t | \mathcal{F}_{t-1})$ of $\mathbf{x}_t$ is the mixture of autoregressive models in (1):

$$\begin{aligned} P(\mathbf{x}_t | \mathcal{F}_{t-1}) &= \sum_{k=1}^K P(\mathbf{x}_t, k_t = k | \mathcal{F}_{t-1}) \\ &= \sum_{k=1}^K \alpha_{k,t} \Phi(\mathbf{x}_t; \mu_{k,t}, \sigma_k^2 I), \end{aligned}$$

which is referred to as the LMVAR model.

Define $\Theta_p = \bigcup_{k=1}^K [\{\beta_k, \sigma_k\} \cup \{\varphi_{k,j} \text{ for } j = 1, \dots, p_k\}]$ and $\Theta_h = \{K, D_1, D_2, q, p_1, \dots, p_K\}$ to be the set of all parameters and the set of all hyperparameters respectively. We estimate Θ_p with the EM algorithm in Section 3.2 and choose Θ_h that gives the best performance in the validation set described in Section 5.3. We define $\hat{\mathbf{x}}_t = \sum_{k=1}^K \alpha_{k,t} \mu_{k,t}$ to be the predicted PCA score. The following section explains how to estimate the model parameters through the EM algorithm.

3.2. Model estimation and forecast

This section estimates the LMVAR model with a training set of size $N: \{\mathbf{x}_t, \mathbf{w}_t | 1 \leq t \leq N\}$. The estimation steps are summarized in Fig. 4(a). Recall that the input of $\mu_{k,t}$ and the function $\mathbf{w}_t$ are the D_1 -dimensional vector $\mathbf{x}_{t-j}$ ($j \in \{1, \dots, p_k\}$) and D_2 -dimensional vector $(w_{1,t}, \dots, w_{D_2,t})$ from the PCA score of the electric load curve, respectively. To maximize complete log-likelihood, we use the EM algorithm in [38]. Let $\mathbf{Z}_t = (Z_{1,t}, \dots, Z_{K,t})$ be a latent vector on day t such that $Z_{k,t} = 1$ if $\mathbf{x}_t$ is generated from pattern k 's distribution and $Z_{k,t} = 0$ otherwise. Then, the complete log-likelihood function is given by

$$\begin{aligned} l &= \sum_{t=p_{\max}+1}^N \sum_{k=1}^K Z_{k,t} \log(\alpha_{k,t} \phi(\mathbf{x}_t | \mu_{k,t}, \sigma_k^2 I)) + \frac{\lambda}{2} \sum_{k=1}^K \|\beta_k\|_2^2 \\ &= \sum_{t=p_{\max}+1}^N \sum_{k=1}^K \left(Z_{k,t} \log \alpha_{k,t} - \frac{n \cdot Z_{k,t}}{2} \log \sigma_k - \frac{Z_{k,t}}{2\sigma_k^2} (\mathbf{x}_t - \mu_{k,t})^T (\mathbf{x}_t - \mu_{k,t}) \right) \\ &\quad + \frac{\lambda}{2} \sum_{k=1}^K \|\beta_k\|_2^2, \end{aligned} \quad (3)$$

where $p_{\max}$ is $\max(p_1, \dots, p_K, q)$, $\phi(\mathbf{x}_t | \mu_{k,t}, \sigma_k^2 I)$ is a probability density function with the mean vector $\mu_{k,t}$ and the covariance matrix $\sigma_k^2 I$, and λ is a regularization parameter. The EM algorithm updates parameters by iterating *E-step* and *M-step*.

In *E-step*, we suppose the parameter set Θ_p is known. The expected value $z_{k,t}$ of the latent variable $Z_{k,t}$ is computed as follows:

For $k \in \{1, \dots, K\}$ and $t \in \{1, \dots, N\}$,

$$\begin{aligned} z_{k,t} &= E(Z_{k,t} | \mathcal{F}_t) \\ &= P(Z_{k,t} = 1 | \mathcal{F}_t) \\ &= \frac{P(Z_{k,t} = 1 | \mathbf{w}_{(t-1):q}) P(\mathbf{x}_t | Z_{k,t} = 1)}{\sum_{j=1}^K P(Z_{j,t} = 1 | \mathbf{w}_{(t-1):q}) P(\mathbf{x}_t | Z_{j,t} = 1)} \\ &= \frac{\alpha_{k,t} \phi(\mathbf{x}_t; \mu_{k,t}, \sigma_k^2 I)}{\sum_{j=1}^K \alpha_{j,t} \phi(\mathbf{x}_t; \mu_{j,t}, \sigma_j^2 I)}. \end{aligned} \quad (4)$$

In *M-step*, the latent variables $Z_{k,t}$ in (3) are replaced with the expected values $z_{k,t}$ in (4). Let l^* denote the replaced complete log-likelihood as follows:

$$\begin{aligned} l^* &= \sum_{t=p_{\max}+1}^N \sum_{k=1}^K \left(Z_{k,t} \log \alpha_{k,t} - \frac{n \cdot Z_{k,t}}{2} \log \sigma_k - \frac{Z_{k,t}}{2\sigma_k^2} (\mathbf{x}_t - \mu_{k,t})^T (\mathbf{x}_t - \mu_{k,t}) \right) \\ &\quad + \frac{\lambda}{2} \sum_{k=1}^K \|\beta_k\|_2^2. \end{aligned} \quad (5)$$

We use the Newton–Raphson method to maximize l^* in (5). The partial derivatives with respect to Θ_p are given by

$$\frac{\partial l^*}{\partial \beta_k} = \sum_{t=p_{\max}+1}^N \left[\left(z_{k,t} - \frac{\exp(\mathbf{w}_{(t-1):q}^T \beta_k)}{\sum_{j=1}^K \exp(\mathbf{w}_{(t-1):q}^T \beta_j)} \right) \cdot \mathbf{w}_{(t-1):q} \right] + \lambda \beta_k, \quad (6)$$

$$\begin{aligned} \frac{\partial l^*}{\partial \beta_k \partial \beta_k^T} &= \sum_{t=p_{\max}+1}^N \frac{\exp(\mathbf{w}_{(t-1):q}^T \beta_k)}{\sum_{j=1}^K \exp(\mathbf{w}_{(t-1):q}^T \beta_j)} \cdot \left(1 - \frac{\exp(\mathbf{w}_{(t-1):q}^T \beta_k)}{\sum_{j=1}^K \exp(\mathbf{w}_{(t-1):q}^T \beta_j)} \right) \\ &\quad \mathbf{w}_{(t-1):q} \mathbf{w}_{(t-1):q}^T + \lambda I, \end{aligned} \quad (7)$$

$$\frac{\partial l^*}{\partial \sigma_k} = \sum_{t=p_{\max}+1}^N \left(-\frac{n \cdot z_{k,t}}{2\sigma_k} + \frac{z_{k,t}}{\sigma_k^3} (\mathbf{x}_t - \mu_{k,t})^T (\mathbf{x}_t - \mu_{k,t}) \right),$$

$$\frac{\partial l^*}{\partial \varphi_{k,j}} = \sum_{t=p_{\max}+1}^N z_{k,t} \sigma_k^{-2} (\mu_{k,t} - \mathbf{x}_t) \mathbf{x}_{t-j}^T,$$

$$\frac{\partial l^*}{\partial \varphi_{k,j} \partial \varphi_{k,j}^T} = \sum_{t=p_{\max}+1}^N z_{k,t} \mathbf{x}_{t-j} \mathbf{x}_{t-j}^T.$$

To help understand (6) and (7), let l^{**} be the expression that consists of the terms including β_k from (5):

$$\begin{aligned} l^{**} &= \sum_{t=p_{\max}+1}^N \sum_{k=1}^K z_{k,t} \log \alpha_{k,t} + \lambda \sum_{k=1}^K \|\beta_k\|_2^2 \\ &= \sum_{t=p_{\max}+1}^N \sum_{j=1}^K z_{j,t} \log \frac{\exp(\mathbf{w}_{(t-1):q}^T \beta_k)}{\sum_{j=1}^K \exp(\mathbf{w}_{(t-1):q}^T \beta_j)} \\ &\quad + \lambda \sum_{k=1}^K \|\beta_k\|_2^2 \\ &= \sum_{t=p_{\max}+1}^N \sum_{k=1}^K z_{k,t} \left(\mathbf{w}_{(t-1):q}^T \beta_k - \log \sum_{j=1}^K \exp(\mathbf{w}_{(t-1):q}^T \beta_j) \right) + \lambda \sum_{k=1}^K \|\beta_k\|_2^2. \end{aligned} \quad (8)$$

We then obtain the first and second derivatives of l^* with respect to β_k by taking the partial derivatives of l^{**} .

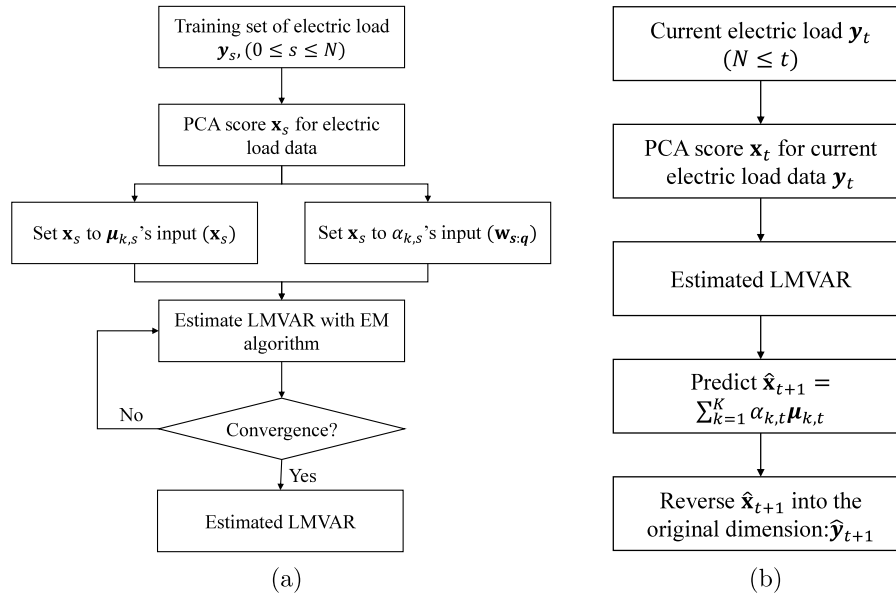


Fig. 4. (a) The LMVAR model estimation (b) Forecasting with the LMVAR.

With the partial derivatives, the parameters are updated as follows:

$$\begin{aligned} \beta_k^{(m+1)} &= \beta_k^{(m)} \\ &- \left(\sum_{t=p_{max}+1}^N \frac{\exp(\mathbf{w}_{(t-1):q}^T \beta_k)}{\sum_{j=1}^K \exp(\mathbf{w}_{(t-1):q}^T \beta_j)} \left(1 - \frac{\exp(\mathbf{w}_{(t-1):q}^T \beta_k)}{\sum_{j=1}^K \exp(\mathbf{w}_{(t-1):q}^T \beta_j)} \right) \right. \\ &\quad \left. \mathbf{w}_{(t-1):q} \mathbf{w}_{(t-1):q}^T + \lambda I \right)^{-1} \cdot \\ &\quad \left(\sum_{t=p_{max}+1}^N \left(z_{k,t} - \frac{\exp(\mathbf{w}_{(t-1):q}^T \beta_k)}{\sum_{j=1}^K \exp(\mathbf{w}_{(t-1):q}^T \beta_j)} \right) \cdot \mathbf{w}_{(t-1):q} + \lambda \beta_k \right), \\ \sigma_k^{(m+1)} &= \sqrt{2 \cdot \frac{\sum_{t=p_{max}+1}^N z_{k,t} (\mathbf{x}_t - \mu_{k,t})^T (\mathbf{x}_t - \mu_{k,t})}{n \sum_{t=p_{max}+1}^N z_{k,t}}}, \\ \varphi_{k,j}^{(m+1)} &= \varphi_{k,j}^{(m)} \\ &- \left(\sum_{t=p_{max}+1}^N z_{k,t} \mathbf{x}_{t-j} \mathbf{x}_{t-j}^T \right)^{-1} \cdot \left(\sum_{t=p_{max}+1}^N z_{k,t} \sigma_k^{-2} (\mu_{k,t} - \mathbf{x}_t) \mathbf{x}_{t-j}^T \right). \end{aligned}$$

Iterating the EM steps gives the estimate of the parameters θ_p when the complete log-likelihood in (5) converges.

Once the model is available, we can forecast the day-ahead electric load curve $\hat{\mathbf{y}}_{t+1}$, following the forecasting procedure in Fig. 4(b). The PCA score of the current electric load $\mathbf{y}_t$ constructs the inputs of the LMVAR model $\mathbf{x}_t$ and $\mathbf{w}_t$. We obtain the predicted PCA score $\hat{\mathbf{x}}_{t+1}$ using the equation, $\hat{\mathbf{x}}_{t+1} = \sum_{k=1}^K \alpha_{k,t} \mu_{k,t}$. Since $\hat{\mathbf{x}}_{t+1}$ is a PCA score, we take the inverse transform to recover the predicted curve $\hat{\mathbf{y}}_{t+1}$.

3.3. Phase variability and curve registration

The LMVAR model has latent variables $Z_{k,t}$ used for determining the load patterns of input curves. As in (4), they are a function of $\alpha_{k,t}$ that is a function of the vector $\mathbf{w}_{t,q}$. That is, the vector $\mathbf{w}_{t,q}$ can contribute to the estimation of $Z_{k,t}$. In Section 2, however, we mention that only electric load data is available, which means that we have no choice but to set $\mathbf{w}_t$ to the D_2 -dimensional PCA score of $\mathbf{y}_t$.

We now think how we can make use of the function $\mathbf{w}_t$ for improving forecasting accuracy instead of setting the value with the PCA score of $\mathbf{y}_t$. We find an idea from curve registration in [34] that aligns curves by removing *phase variability*. Fig. 5 from [34] is a good example for explaining phase variability. When we see Fig. 5(a), we do not see

any patterns in the curves. After aligning them by scaling and shifting with warping functions in Fig. 5(c), we then see the characteristics of curves having five peaks, as in Fig. 5(b). *Phase variability* of curves is such variability along the time or horizontal axis that hides the characteristics of curves.

We, therefore, expect some improvement in the clustering of curves from curve registration. We will explain how we apply curve registration to the proposed model with a brief introduction to the curve registration in the following section.

4. LMVAR with curve registration

Curve registration in [34] and [35] aligns curves horizontally using optimization for reducing phase variability. We apply curve registration to the electric load curves $\mathbf{y}_t$ before performing dimension reduction. Briefly explaining curve registration, it aligns square-root velocity functions (SRVFs) of curves denoted by $q(\cdot)$. For a given curve $\zeta(\cdot)$, the function $q(\cdot)$ maps $[0, 1]$ into $\mathbb{R}$. The definition of $q(\cdot)$ is given by:

$$q(h) \equiv Q(\zeta(h)) = \zeta(h) / \sqrt{|\dot{\zeta}(h)|},$$

where $\dot{\zeta}(\cdot)$ is a first-order derivative of $\zeta(\cdot)$.

The algorithm computes the Karcher mean of the orbits of SRVFs as a template. Then, it finds a warping function for each curve, aligning it to the Karcher mean. The electric load curves transform into the warped curves with their warping functions. We refer the reader to [34] and [35] for the details of curve registration.

Let $\mathbf{y}'_t = (y'_{1,t}, y'_{2,t}, \dots, y'_{H,t})$ be the warped curve and γ_t be its warping function on day t . Fig. 6(a) presents daily electric load curves $\mathbf{y}_t$ in a university library. Curve registration constructs warped curves $\mathbf{y}'_t$ as in Fig. 6(b) using warping functions γ_t in Fig. 6(c). We observe that the curves are aligned at jump points around time 30.

We use warped curves for creating the vector $\mathbf{w}_{(t-1):q}$ by setting the function $\mathbf{w}_t$ with the PCA scores of the warped curves. Recall that $\mathbf{w}_{(t-1):q}$ is used to determine the latent variables $Z_{k,t}$, the load pattern of input curves. We, therefore, anticipate improvement in internal load patterning of the model. We call this approach the LMVAR with curve registration (LMVAR-CR) and explain its estimation and forecasting in Section 4.1.

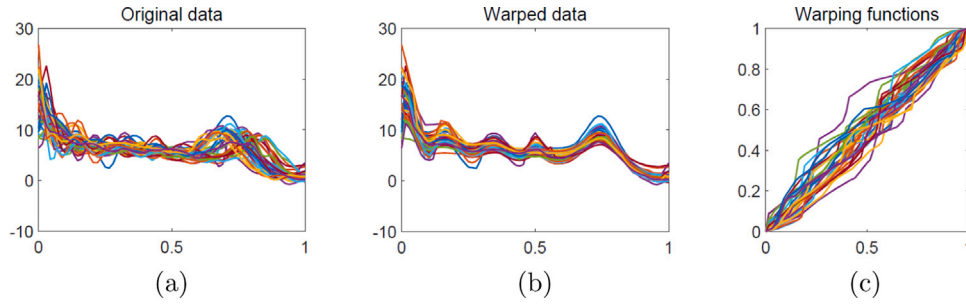


Fig. 5. (a) Original growth data for 39 males (b) Warped growth curves (c) Warping functions of warped growth curves, [34].

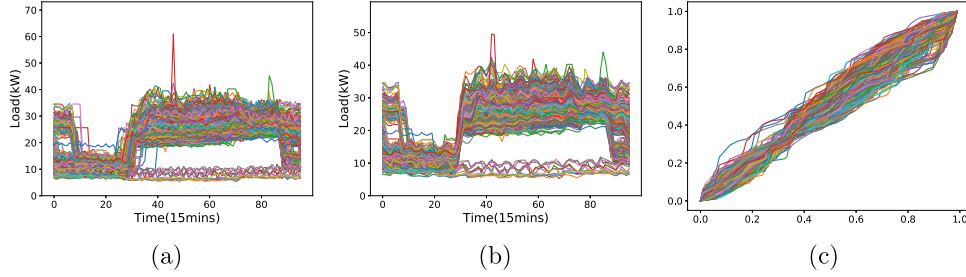


Fig. 6. (a) Electric load curves for the library (b) Warped electric load curves for the library (c) Warping functions for the library.

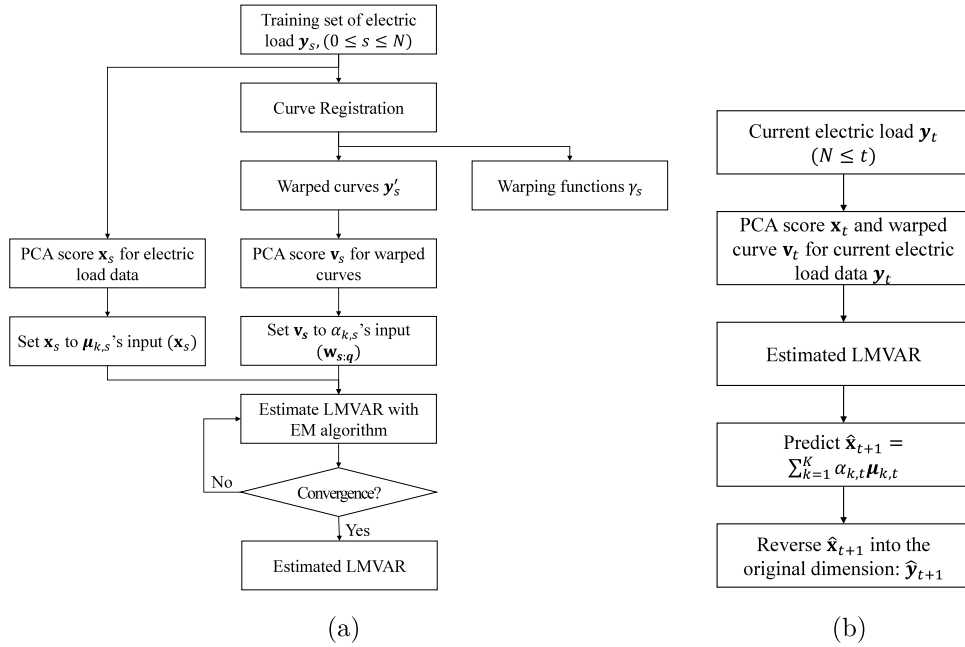


Fig. 7. (a) The LMVAR-CR model estimation (b) Forecasting with the LMVAR-CR.

4.1. LMVAR-CR estimation and forecasting

Fig. 7(a) shows the estimation procedure with the LMVAR-CR model. Unlike the LMVAR model, the LMVAR-CR model uses warped curves to construct w_t . For $s \in \{1, \dots, N\}$, curve registration aligns the electric load curves y_s and generates the warped curves with warping functions. Let v_s be the D_2 -dimensional PCA score of a warped curve y'_s at time s . We construct x_s and v_s using the PCA scores of y_s and y'_s respectively. As in the LMVAR model, x_s are the input of the VAR term $\mu_{k,s}$. We set $w_s = v_s$ and use it as the input of the mixing proportions $\alpha_{k,s}$. Then, the EM algorithm in Section 3.1 iterates the estimates until it converges.

The forecasting procedure in Fig. 7(b) is the same as that of the LMVAR except that we use v_t for the vector $w_{t:q}$.

5. Numerical studies

We use evaluate our two proposed models (LMVAR and LMVAR-CR) using the nested cross-validation for time series data (Fig. 8). The available data splits into eight data subsets. The first four data subsets are a training set, and the fifth data subset is a test set. Each iteration increases the training set by one subset with the test set moving to the next subset.

We use four popular performance measures: root mean squared error (RMSE), forecast skill (FS), coefficient of variance (CV), and mean

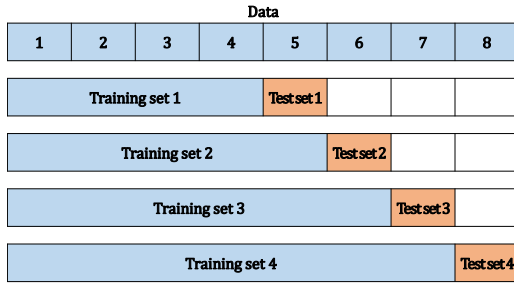


Fig. 8. Nested cross-validation.

absolute percentage error (MAPE). Let $y_{h,t}$ denote the electric load measured at time h of each day t in the test set, and let $\hat{y}_{h,t}$ represent the corresponding prediction. The overall RMSE is defined as

$$\text{RMSE} = \sqrt{\frac{1}{N_{\text{test}} \cdot H} \sum_{t=1}^{N_{\text{test}}} \sum_{h=1}^H (y_{h,t} - \hat{y}_{h,t})^2},$$

where N_{test} is the size of the test set, and H is the number of measurements on a day. The RMSE at a specific time h of day is

$$\text{RMSE}_h = \sqrt{\frac{1}{N_{\text{test}}} \sum_{t=1}^{N_{\text{test}}} (y_{h,t} - \hat{y}_{h,t})^2} \text{ for } h = 1 \dots, H.$$

The FS, CV and MAPE are defined as

$$\text{FS} = \left(1 - \left(\frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{persistence}}} \right)^2 \right) \times 100,$$

$$\text{CV} = \frac{\sqrt{\frac{1}{N_{\text{test}} \cdot H - 1} \sum_{t=1}^{N_{\text{test}}} \sum_{h=1}^H (y_{h,t} - \hat{y}_{h,t})^2}}{\bar{y}} \times 100,$$

$$\text{MAPE} = \frac{1}{N_{\text{test}} \cdot H} \sum_{t=1}^{N_{\text{test}}} \sum_{h=1}^H \left| \frac{y_{h,t} - \hat{y}_{h,t}}{y_{h,t}} \right| \times 100,$$

where $\bar{y}$, $\text{RMSE}_{\text{model}}$ and $\text{RMSE}_{\text{persistence}}$ are the mean of $y_{h,t}$, the RMSEs of the model and persistence respectively.

5.1. Data description

The proposed and benchmark models use real historical electric load (kW) data from two different buildings described in Table 1. The one data set is from a university library building located in Pohang city, Republic of Korea, between January 26, 2018 and December 12, 2019. The other is a grocery building data set collected from January 1, 2012 to December 31, 2012, publicly available from the EnerNOC Commercial building dataset [39]. The library and grocery data are available for 685 and 365 days, respectively, excluding the days having no measurements. The library electric loads were measured every 15 min; the number of measurements per day is 96. The day-ahead forecasting predicts 96 electric loads for the next day, based on all past electric loads. The grocery data were collected every 5 min, and the number of measurements per day is 288.

The noticeable difference between the library and grocery is that the daily load for the library has unique four patterns, as in Fig. 1(b), but the grocery data does not show noticeable patterns by jumps, as shown in Fig. 9. We think that testing with data sets with/without distinct patterns will partially prove our models' general applicability.

5.2. Benchmark models

We compare our models against six benchmark models: persistence, seasonal autoregressive (SAR), multi-output support vector machine (M-SVM), multilayer perceptron (MLP), S2S LSTM, and a cluster-based model.

Table 1

Building load data set characteristics.

Building	Library	Grocery
Location	Pohang, Republic of Korea	New York, USA
H (min)	96 (15)	288 (5)
Min load (kW)	5.272	2.253
Mean load (kW)	21.21	25.90
Max load (kW)	117.78	56.65
std (kW)	7.890	6.282

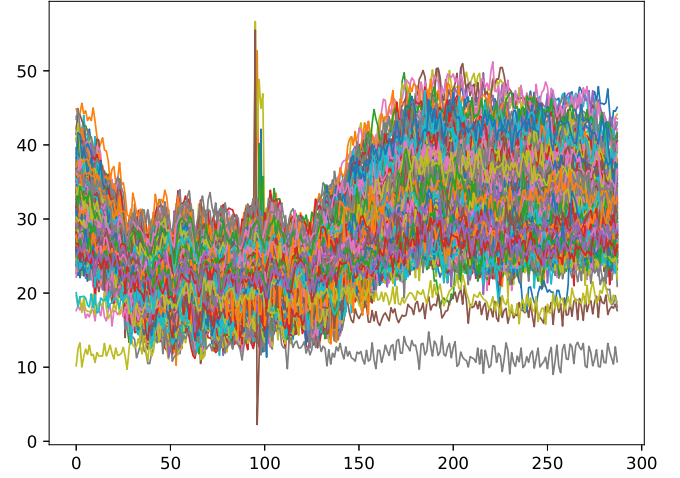


Fig. 9. Daily electric load profiles for the grocery building.

Persistence

Persistence is a simple predictive model where the electric load forecast for the next day is equal to the electric load on the current day as follows [25]:

$$\hat{y}_{t+1} = y_t.$$

Seasonal autoregressive (SAR)

The standard SAR model takes into account seasonality of existing autoregressive with non-seasonal time lag p^{sar} , seasonal time lag P^{sar} and the period H . The SAR model is defined as,

$$\left(1 - \sum_{j=1}^{p^{sar}} a_j B^j \right) \left(1 - \sum_{j=1}^{P^{sar}} b_j B^{H \cdot j} \right) y_{t'} = \epsilon_{t'},$$

where $\{y_{t'}\}_{t'=1}$ is an electric load data as a univariate time series, B^j is a backshift operation at lag j , a_j and b_j are the coefficients of the model, and $\epsilon_{t'} \sim \Phi(\epsilon_{t'}|0, \sigma_\epsilon)$ is noise term. In the numerical studies, both time lags p^{sar} and P^{sar} are set to be one, the same as the proposed LMVAR model.

Multi-output support vector machine (M-SVM)

Pérez-Cruz et al. [40] proposed the multi-output support vector machine (M-SVM) as the generalized version of the standard SVM, and Bao et al. [32] showed that M-SVM works well for time series forecasting.

Given the previous electric load curve y_t , the M-SVR predicts the next day load curve as follows,

$$\hat{y}_{t+1} = W\psi(y_t) + \mathbf{b}$$

where W is a $H \times N$ weight matrix, $\mathbf{b}$ is a H -dimensional bias vector, and $\psi(\cdot)$ is a nonlinear transformation such that the kernel K satisfies that $K(y_{t_1}, y_{t_2}) = \psi(y_{t_1})^T \psi(y_{t_2})$. In the numerical studies, we use a radial basis function (RBF) kernel, $K(y_{t_1}, y_{t_2}) = \exp\left(-\frac{\|y_{t_1} - y_{t_2}\|_2^2}{2\eta^2}\right)$ where η

Table 2
Hyperparameters over four validation tests for the library building.

Model (Hyperparameters)	Test 1	Test 2	Test 3	Test 4
M-SVM (C, η)	(100, 0.0001)	(100, 0.0001)	(1000, 0.001)	(1000, 0.001)
MLP (Hidden units)	50	50	500	100
S2S LSTM (Hidden units)	15	50	50	50
Cluster model (K)	4	4	4	4
LMVAR (K, D_1, D_2)	(4, 5, 2)	(4, 4, 3)	(4, 5, 2)	(4, 3, 2)
LMVAR-CR (K, D_1, D_2)	(4, 6, 2)	(4, 4, 3)	(4, 6, 2)	(4, 5, 2)

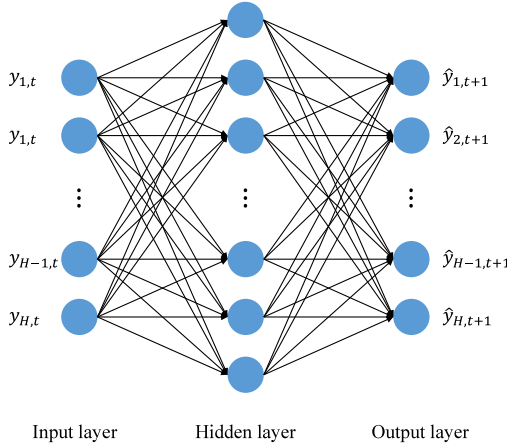


Fig. 10. Multilayer perceptron for a day-ahead electric load forecast.

is a scale hyperparameter. The M-SVM model optimizes the following problem to obtain the weight matrix W and $\mathbf{b}$,

$$L(W, \mathbf{b}) = \frac{1}{2} \|W\|_F^2 + C \sum_{i=1}^N L(u_i),$$

where $u_i = \|\mathbf{e}_i\| = \sqrt{\mathbf{e}_i^T \mathbf{e}_i}$, $\mathbf{e}_i = y_i - W\psi(y_{i-1}) - \mathbf{b}$, C is a hyperparameter, and $L(u) = \begin{cases} 0 & u < \epsilon \\ u^2 - 2u\epsilon + \epsilon^2 & u \leq \epsilon \end{cases}$. We find the optimal solution using iterative reweighted least squares (IRWLS) procedure in [41].

Multilayer perceptron (MLP)

Kim et al. [31] compared a multilayer perceptron (MLP) as a benchmark model for electric load forecasting in Fig. 10. This MLP is a multi-input multi-output model with one hidden layer as the simple model of an artificial neural network. The input is the previous day load curve y_t , and the output is the next day load curve $\hat{y}_{t+1}$.

Sequence to sequence LSTM (S2S LSTM)

Marino et al. [21] applied sequence to sequence LSTM (S2S LSTM) for building energy load forecasting. As shown in Fig. 11, S2S LSTM is composed of encoder LSTM and decoder LSTM. Encoder LSTM takes loads and timestamp ($1 \sim H$) as an input and gives the hidden state to decoder LSTM. Decoder LSTM obtains the next day timestamp ($1 \sim H$) as an input and predicts the next day loads $\hat{y}_{t+1}$.

Cluster model

The cluster-based model in [25] is to cluster daily electric loads and construct transition probabilities between load patterns of days. It also predicts the total amount of the electric load of a day. Then, it chooses the centroid of the curves in the load pattern, having the highest transition probability. It finally forecasts a day-ahead electric curve by scaling the centroid curve with the predicted total electric load.

5.3. Parameter initialization

We notice that parameter initialization has a significant effect on model performance. Especially, we need to initialize the latent variables $\mathbf{Z}_t$ as well as the parameters Θ_p . The number of load patterns K can be chosen, and the latent variables $\mathbf{Z}_t$ can be initialized with a clustering algorithm simultaneously. For our data set, it is not difficult to find that the PCA scores $\mathbf{w}_t$ come from a mixture of distributions. Therefore, this paper chooses the number of clusters that has minimum Silhouette Coefficient score [42]. As shown in Fig. 12, the library case gets $K = 4$ clusters. For the parameters of the vector autoregressive terms, each VAR's coefficient $\phi_{k,1}$ is initialized to an identity matrix, and $\phi_{k,j}$ for each $j = 2 \dots, p_k$ is set to a zero matrix. Finally, we set σ_k to one.

The model selection determines hyperparameters in advance. As shown in Fig. 13, we split the training set into a training subset (80%) and a validation subset (20%). We estimate the model with the training subset and choose the hyperparameters when the model has the best RMSE on the validation subset. With those hyperparameters, we estimate the model with the training set and evaluate it with the test set.

From the model selection, we have the hyperparameters including the time lags of p_k and q , and the dimensions D_1 and D_2 . The values that we consider for hyperparameters are $p_k \in \{1, \dots, 7\}$, $q \in \{1, \dots, 7\}$, $D_1 \in \{2, \dots, 6\}$ and $D_2 \in \{2, 3\}$ with $p_1 = \dots = p_K = q$. We determine the hyperparameters from $7 \cdot 5 \cdot 2 = 70$ possible settings. Similarly, we choose the set of hidden units in MLP from $\{5, 10, 20, 50, 100, 200, 500\}$ and in S2S LSTM from $\{5, 10, 15, 20, 30, 50, 100\}$. The hyperparameters C and η of MLP are from $\{0.1, 1, 10, 100, 1000\}$ and $\{1, 10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}\}$, respectively. The cluster model chooses the number of clusters using Silhouette Coefficient score [42]. The settings of SAR's time lags are with $p^{sar} = 1$ and $p^{sar} = 1$.

5.4. Case study 1: The library electric load forecasting

This section explains the experimental result for the library data. We write and run code under a regular PC environment with Intel(R) Core(TM) i7-4790 @ 3.60 GHz CPU and 16GB RAM. Curve registration was done with MATLAB R2017b. We use sklearn library [43] in Python for the parameter initialization. The LMVAR models are written in Python.

Table 2 explains the hyperparameters of two proposed models and benchmark models over four tests of the library data in the nested cross validation. We do not see any specific relations of D_1 and D_2 across four tests. The model selection gives $p_1 = \dots = p_K = q = 1$ for all the tests and models.

Table 3 shows the RMSEs of the proposed and benchmark models on four test sets. We see that the SAR has the worst average RMSEs (5.388). The S2S LSTM and the MLP follow the SAR (4.999 and 5.150, respectively). As the training set size increases, the RMSEs of S2S LSTM tend to decrease gradually. The MLP shows the lower RMSE value in the first test out of four tests, but does not have consistent RMSE values. The M-SVM is better than the SAR, the MLP and the S2S LSTM, especially in the second test. However, the M-SVM does not beat persistence on average. The cluster model is slightly better than persistence for all test sets. We observe that the proposed models generally perform better than other benchmark methods. For test 1, the

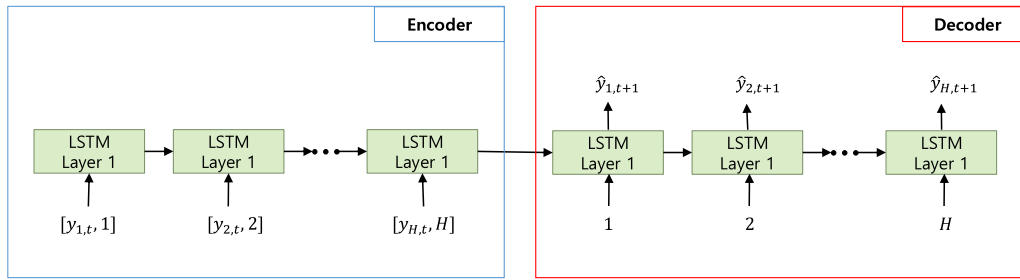


Fig. 11. Sequence to sequence LSTM for day-ahead electric load forecast.

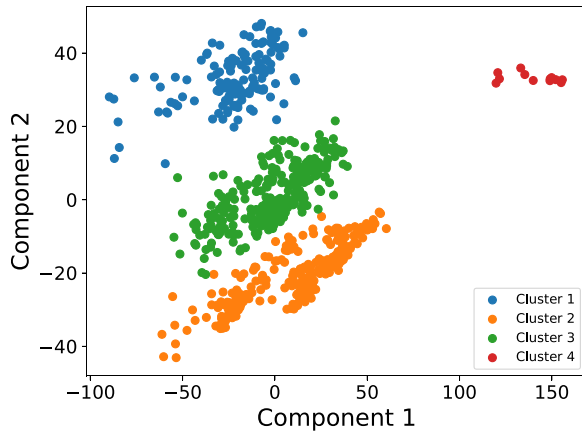


Fig. 12. PCA score for the library electric load curves.

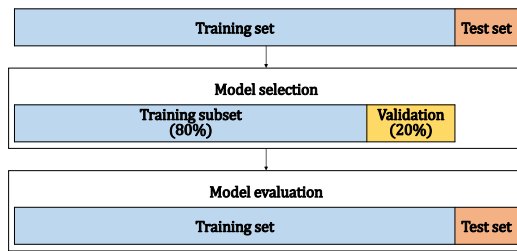


Fig. 13. Model selection and evaluation.

Table 3
RMSE of forecasting models for the library.

	Test 1	Test 2	Test 3	Test 4	Average
Persistence	3.114	3.010	2.777	2.904	2.951
SAR	3.540	6.617	6.119	4.693	5.388
M-SVM	2.849	2.496	3.485	3.214	3.031
MLP	3.751	6.660	4.457	5.232	5.150
S2S LSTM	5.631	5.129	5.147	4.090	4.999
Cluster model	3.059	2.811	2.672	2.754	2.824
LMVAR	2.734	2.673	2.872	2.584	2.716
LMVAR-CR	2.934	2.762	2.461	2.489	2.661

LMVAR model gives better RMSE than the LMVAR-CR model. On the other hand, the LMVAR-CR model has the lowest RMSEs for tests 3 and 4 and has the best average RMSE, followed by the LMVAR model.

Comparing the models with persistence as a baseline, Table 4 presents the FSs of the forecast models. The SAR gives the worst average FS (-233%), followed by the MLP and the S2S LSTM. In contrast to the RMSEs, the FSs of the S2S LSTM do not improve significantly compared to persistence, even if the training set size increases. the M-SVM performs better than persistence in tests 1 and 2, but FSs of the M-SVM are degraded in tests 3 and 4. The LMVAR model gives good FS

Table 4
FS (%) of forecasting models for the library.

	Test 1	Test 2	Test 3	Test 4	Average
Persistence	0	0	0	0	0
SAR	-29.21	-383	-385	-161	-233
MLP	-45.09	-389	-157	-224	-204
M-SVM	16.29	31.27	-57.53	-22.42	-5.471
S2S LSTM	-226	-190	-243	-98	-186
Cluster model	3.50	12.81	7.43	10.10	8.45
LMVAR	22.90	21.16	-6.97	20.83	15.32
LMVAR-CR	11.19	15.83	21.46	26.54	18.67

Table 5
CV of forecasting models for the library.

	Test 1	Test 2	Test 3	Test 4	Average
Persistence	16.22	12.51	11.99	12.98	13.31
SAR	18.48	27.55	26.35	21.17	24.33
MLP	19.59	27.73	19.19	23.60	23.25
M-SVM	14.88	10.39	15.01	14.50	13.68
S2S LSTM	29.16	21.33	22.84	18.07	22.73
Cluster model	15.93	11.68	11.54	12.31	12.74
LMVAR	14.24	11.61	11.29	11.55	12.09
LMVAR-CR	15.29	11.47	10.63	11.13	12.02

Table 6
MAPE of forecasting models for the library.

	Test 1	Test 2	Test 3	Test 4	Average
Persistence	9.280	8.756	8.350	8.522	8.727
SAR	13.16	25.15	19.16	17.10	18.68
MLP	18.69	21.46	14.96	19.92	18.77
M-SVM	10.86	8.881	11.48	11.11	10.57
S2S LSTM	34.16	15.78	20.73	15.97	21.66
Cluster model	10.89	9.221	9.088	8.678	9.470
LMVAR	10.42	9.199	9.278	9.503	9.600
LMVAR-CR	11.14	9.074	8.325	8.699	9.312

for tests 1, 2, and 4, but does not perform well for test 3. The LMVAR-CR model becomes better as the training set size increases and gives the best average FS.

The CV in Table 5 shows the variability of difference between real and predicted values to the load mean. The SAR gives the worst average CV (24.33), followed by the MLP and the S2S LSTM. The M-SVM is the best CV (10.39) in test 2, and the LMVAR model works well in test 1. The LMVAR-CR model gives good CVs in all tests.

The MAPE in Table 6 is the relative error rate of the actual load. Unlike other measures, persistence gives the best MAPEs in most tests. It means that the LMVAR and the LMVAR-CR models are good at predicting the bad cases, but on average, persistence works well.

To see the actual forecasting trace, Fig. 14 plots weeklong trajectories of true load data and forecasts of models from September 23, 2019 (Monday) through September 29, 2019 (Sunday). For clear visualization, Figs. 14 and 15 plot select models that are representative and perform relatively well. The S2S LSTM constructs similar-shaped curves for seven days, as the input of the S2S LSTM is the electric load of the previous day. The LMVAR(-CR) and the cluster models depend

Table 7

Hyperparameters over four validation tests for the grocery building.

Model (Hyperparameters)	Test 1	Test 2	Test 3	Test 4
M-SVM (C, η)	(100, 0.01)	(1, 0.0001)	(1, 0.0001)	(10, 0.0001)
MLP (Hidden units)	5	500	20	5
S2S LSTM (Hidden units)	100	50	100	5
Cluster model (K)	3	3	3	3
LMVAR (K, D_1, D_2)	(3, 2, 2)	(3, 4, 2)	(3, 3, 3)	(3, 5, 2)
LMVAR-CR (K, D_1, D_2)	(3, 4, 3)	(3, 4, 3)	(3, 5, 3)	(3, 6, 2)

Table 8

RMSE of forecasting models for the grocery building.

	Test 1	Test 2	Test 3	Test 4	Average
Persistence	4.053	3.752	2.573	2.539	3.301
SAR	9.756	7.625	4.994	4.158	4.158
MLP	11.30	7.177	3.933	3.858	7.239
M-SVM	20.65	3.537	3.009	3.072	10.69
S2S LSTM	10.66	5.115	3.812	3.526	6.457
Cluster model	4.553	4.380	3.850	2.513	3.907
LMVAR	3.793	3.173	2.681	2.199	3.020
LMVAR-CR	3.725	3.213	2.211	2.205	2.913

on the previous day. The cluster model predicts the electric load pattern and scales it with the total load of the previous day. The LMVAR-CR model uses the PCA score of the previous day's load curve. Though it is not obvious in Fig. 14, the cluster model works best among the benchmark methods, but the proposed LMVAR-CR model outperforms the cluster model. The reason is that the cluster model uses the centroid of the curves in a pattern; the predicted curves have the same shape (centroid). So, it cannot reflect the variation within a pattern. On the other hand, the LMVAR-CR considers the variation from the recent trends and makes a forecast by modifying the shape within the same load pattern.

The changes in the total load between weekdays and weekends cause to increase forecasting errors. For example, the total load on Friday ($h = [385, 480]$) is higher than that on Saturday ($h = [385, 480]$). Similarly, the Sunday–Monday difference causes forecasting errors.

Fig. 15 shows the $RMSE_h$ over time—we recall the definition of $RMSE_h$ at the beginning of Section 5. The S2S LSTM has the highest $RMSE_h$ for almost all h , and the LMVAR-CR model has the lowest $RMSE_h$ for almost all h . We observe that there is quite a difference between the LMVAR-CR and the cluster model around $h = 72$ and thereafter. The electric load has a large variation at the late time of a day (See Fig. 6(a)). As mentioned earlier, the cluster model weakens the ability to deal with the variation within a pattern and gives a higher $RMSE_h$ than LMVAR-CR at those times. Interestingly, all models commonly have a peak $RMSE_h$ when $h \in [28, 32]$, which is between 7:00 am and 8:00 am. The peak error occurs when the electric load of the building suddenly increases (jumps). It turns out to be difficult to predict the moment of a jump. That is why the $RMSE_h$ is greater than elsewhere.

5.5. Case study 2: The grocery building electric load forecasting

We conduct additional test to make sure that the proposed models work well even if the electric load data does not have distinct patterns separated by jumps. As shown in Table 7, the number of cluster is 3 when Silhouette Coefficient score gives the minimum value. The other hyperparameters are chosen by nested cross-validation.

The results for the grocery data show that the proposed models outperform benchmark models. In Table 8, the M-SVM is the worst in test 1, and the average RMSE of the M-SVM is the highest. Although the cluster model works well for the library data, it does not for the grocery data. The LMVAR and the LMVAR-CR models show remarkable performance compared with other models.

In terms of ratio, we can see more clearly that performance has improved in Table 9. All models except the proposed models are

Table 9

FS of forecasting models for the grocery building.

	Test 1	Test 2	Test 3	Test 4	Average
Persistence	0	0	0	0	0
SAR	−479	−312	−276	−168	−348
MLP	−677	−265	−133	−130	−380
M-SVM	−2495	11.15	−36.77	−46.35	−949
S2S LSTM	−591	−85.81	−119	−92.77	−282
Cluster model	−26.14	−36.24	−123	2.039	−40.08
LMVAR	12.44	28.49	−8.569	24.98	16.29
LMVAR-CR	15.54	26.68	26.17	24.61	22.09

Table 10

CV of forecasting models for the grocery building.

	Test 1	Test 2	Test 3	Test 4	Average
Persistence	12.38	13.05	9.685	10.45	11.75
SAR	29.81	26.53	18.79	17.12	24.90
MLP	34.53	24.97	14.80	15.88	25.78
M-SVM	63.11	12.30	11.32	12.65	38.09
S2S LSTM	32.57	17.79	14.35	14.52	22.99
Cluster model	13.91	15.26	14.51	10.37	13.92
LMVAR	11.58	11.03	10.07	9.048	10.74
LMVAR-CR	11.37	11.17	8.311	9.070	10.36

Table 11

MAPE of forecasting models for the grocery building.

	Test 1	Test 2	Test 3	Test 4	Average
Persistence	9.320	10.13	7.423	7.575	8.613
SAR	28.01	23.55	16.12	14.53	20.55
MLP	31.11	20.71	11.92	13.03	19.19
M-SVM	55.57	9.491	9.561	10.49	21.28
S2S LSTM	29.20	13.48	11.76	12.75	16.80
Cluster model	11.01	12.22	11.78	7.894	10.72
LMVAR	9.102	8.445	8.232	6.773	8.138
LMVAR-CR	8.642	8.429	6.469	6.791	7.583

worse than persistence. Especially, only the LMVAR-CR model gives all positive FSs in all tests.

In Table 10, the SAR, the MLP, the M-SVM, and the S2S LSTM show higher CV values for the grocery data than those for the library data. On the other hand, persistence, the LMVAR, and the LMVAR-CR have lower CV values for the grocery data than those for the library data. The LMVAR-CR model has the lowest average CV (10.36).

The MAPE in Table 11 varies considerably from the library case. The LMVAR and LMVAR-CR for the grocery data give better MAPEs than persistence. These results of RMSE and MAPE imply that the proposed models do forecast well whether the error is weighted according to its size or not.

Fig. 16 shows the prediction values of the M-SVM, the S2S LSTM, the cluster model, and the LMVAR-CR for test 2 of the grocery data. Similar to the library case, the S2S LSTM predicts that the forecast curves are always similar and lower than the actual loads. The common characteristic is that all models produce smoother curves than the actual loads, so it is hard to predict the curve's valleys like the first and third days in Fig. 16.

The $RMSE_h$ s for the overall tests are illustrated in Fig. 17. The M-SVR gives the high $RMSE_h$ because it performs poorly in test 1. The $RMSE_h$ suddenly increases at $h \in [90, 95]$, and that time is the time when the electric load increases rapidly. The LMVAR, the LMVAR-CR, and the cluster model give low RMSEs through the overall forecasting horizon.

Summarizing the numerical studies, we evaluate the proposed models with RMSE, FS, CV, and MAPE using the library and grocery data sets. Benchmark models include the MLP and the SAR in time series, the S2S LSTM in neural networks, the M-SVM in machine learning, and the cluster model. We use nested cross-validation as a validation method. We divide a training dataset into training and validation subsets and

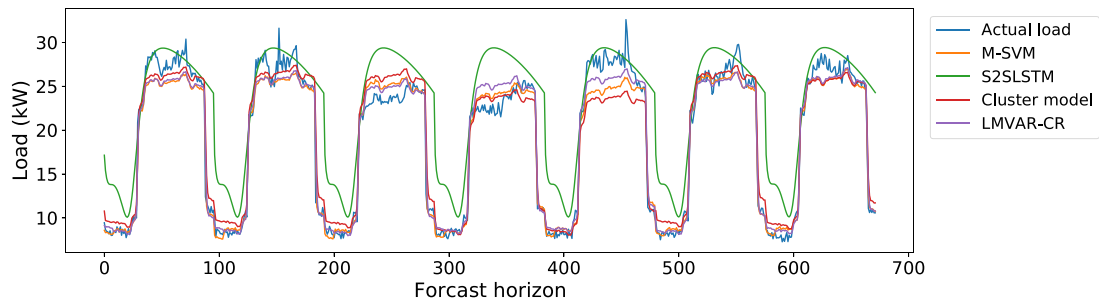


Fig. 14. Comparing the forecasting models for the library data.

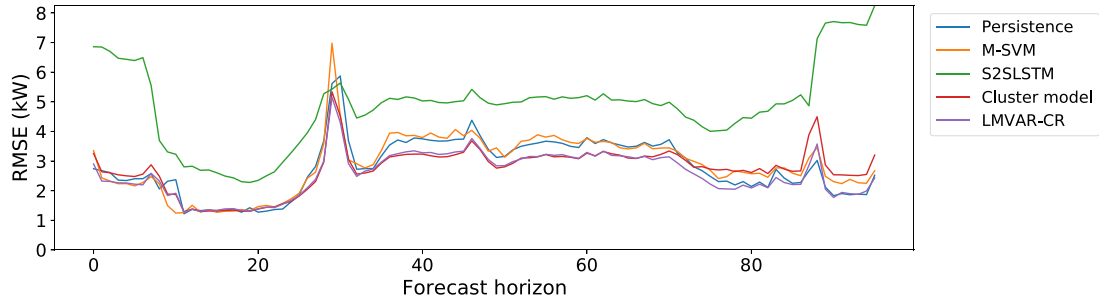
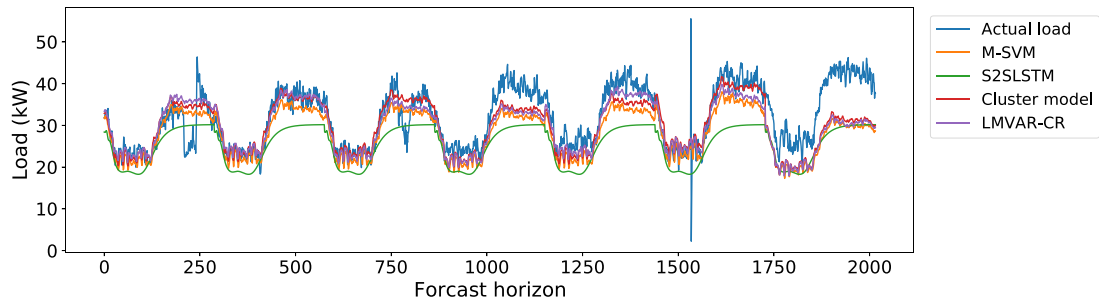
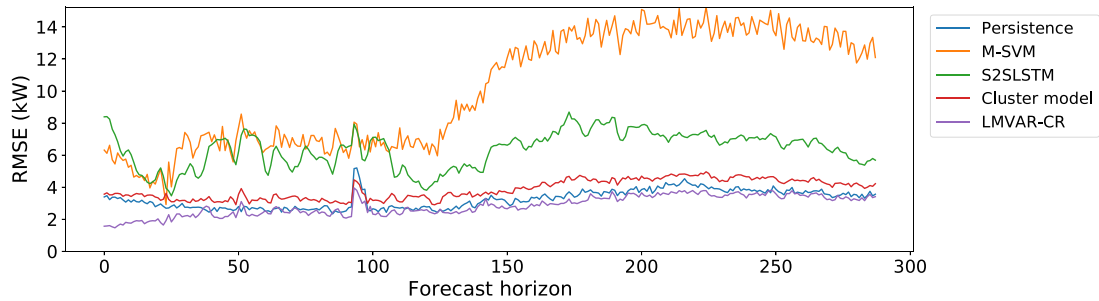
Fig. 15. $RMSE_h$ for forecast horizon h for the library data.

Fig. 16. Comparing the forecast models for the grocery data.

Fig. 17. $RMSE_h$ for forecast horizon h for the grocery data.

choose the hyperparameters when the models with them give the best measure like RMSE for the validation subset.

The proposed models outperform other benchmark models for RMSE, FS, and CV for the library data. Persistence, however, gives better MAPE than the proposed models. It seems that the LMVAR and the LMVAR-CR forecast the worst-case well (better RMSE and CV).

For the grocery data, the proposed models show outstanding performance for all measures. The LMVAR-CR model indeed outperforms all other models in all performance measures.

It is noteworthy to mention that the LMVAR-CR model shows excellent results in most tests. It implies that curve registration has a positive effect on improving forecasting accuracy.

6. Conclusion

This paper proposes the LMVAR and LMVAR-CR models for day-ahead electric load forecasting with only electric load data when daily load presents some patterns and jumps. We assume that each pattern

has an autoregressive model and use the LMVAR model as the mixture of vector autoregressive models. While dealing with vector-valued inputs, we use the PCA decomposition for dimension reduction and transform electric load data into the PCA score. The LMVAR model is estimated with the EM algorithm. The LMVAR-CR model considers phase variability with curve registration to improve internal clustering; the LMVAR-CR reduces the phase variability with curve registration and obtains more distinct patterns. The LMVAR-CR follows the same procedure as the LMVAR except that the input of mixing proportion is the PCA score of warped data of curve registration. We compare those models with existing methods such as persistence, the S2S LSTM, the SAR, the M-SVM, the MLP, and the cluster model.

We test the proposed models and benchmark models using the nested cross-validation. The result indicates that the LMVAR-CR model shows the best performance among all other methods. The fact that the LMVAR-CR outperforms the LMVAR implies that applying curve registration is effective.

There will be some future work extending the proposed model. First, one can incorporate other environmental factors such as weather conditions into the model for the load data highly dependent on environments. Second, weekly patterns (higher load on weekdays and lower on weekends) can be considered in the model appropriately to improve forecasting accuracy. Finally, improving the prediction of jumps in load curves and pattern changes can be a possible extension.

CRedit authorship contribution statement

Dongyeon Jeong: Conceptualization, Methodology, Software, Formal analysis, Writing - original draft. **Chiwoo Park:** Conceptualization, Methodology, Supervision. **Young Myoung Ko:** Conceptualization, Methodology, Writing - review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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